FADA: Feature Aligned Domain Adaptive Object Detection in Remote Sensing Imagery

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***Abstract*— Deep learning-based object detectors have been widely adopted in the field of remote sensing imagery interpretation. These detectors heavily depend on the expensive large-scale labeled datasets, while the scarce remote sensing datasets limit the performance. The domain adaptive object detection can alleviate this problem. However, it struggles with the confusing feature's alignment, damaging the domain generalization performance, especially for the remote sensing scene with sparse objects and diverse backgrounds. For that reason, a semisynthetic data generator (SDG) is proposed to automatically generate the remote sensing dataset with low cost and replace the real-world training dataset, a *feature aligned domain adaptive object detector* (FADA) is proposed to enhance the domain adaptation among the cross-domain remote sensing images. The FADA contains two proposed modules in addition to the base detector: an adversarial-based foreground alignment (AFA) and a prototype-based confusing feature alignment (PCFA). The AFA aligns the cross-domain foreground feature by adversarial training (AT), and it can filter the noisy background feature that is not suitable to transfer. Then, the PCFA adaptively aligns the confusing background and foreground feature, further promoting the domain adaptation performance. Comprehensive experiments validate the effectiveness of the proposed method. Compared with the baseline model trained on the semisynthetic source dataset, our FADA improves the generalized performance on the real-world target dataset a large-scale Dataset for Object deTection in Aerial images (DOTA) by 15.7% average precision (AP) and achieves state-of-the-art results.**

***Index Terms*— Domain adaptation, feature alignment, object detection, remote sensing imagery.**

I. Introduction

REMOTE sensing imagery interpretation has been advanced in the past decade. In this field, detecting the geospatial object in aerial imagery is a hot-spot subject and has been widely studied and continuously developed.

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Recently, the deep learning-based object detectors [2]–[7] have shown impressive performance and have been adopted into varied remote sensing scenes. These data-driven detectors [8]–[12] are under the supervised end-to-end framework. Thus, they are heavily dependent on large-scale labeled datasets, such as a large-scale Dataset for Object deTection in Aerial images (DOTA) [13] and benchmark for object DetectIon in Optical Remote sensing images (DIOR) [14]. To relieve the demand for the labeled data, few-shot learning for aerial object detection [15]–[19] is studied and has achieved promising results. However, these methods can hardly deal with the dataset that has a large domain gap against the few support samples [16]–[18] when the data source changes a lot.

Correspondingly, the study of domain adaptive object detection [20] provides another promising solution. It can extract the knowledge of object detection from the source dataset and transfer it to the target dataset in an unsupervised manner; hence, the labeled samples of the target dataset are not necessary anymore. These two datasets are generally drawn from two similar but not identical distributions due to diverse data sources (such as two satellites with different sensors, resolutions, and bands). The difference between these distributions is widely named the domain gap; domain adaptation aims to bridge this gap and improve the generalization performance on the target dataset.

In remote sensing, most of the domain adaptation works focus on semantic segmentation and image classification [21], [22], but little effort is made to investigate the domain adaptive object detection. In the past few years, the study of domain adaptive object detection has attracted researchers' attention. Many detectors [20], [23], [24] are proposed in the standard computer vision field. Even though these detectors have the potential to be adopted into the remote sensing scene, the domain adaptation among remote sensing images is still challenging as the following two aspects.

1) Expensive Annotated Dataset: In some remote sensing object detection tasks, well-annotated datasets are rare and expensive to acquire. The remote sensing object is sparse, and the targets in a shot of imagery are of few numbers; hence, a large-scale dataset usually consists of many images from different satellites, multitemporal, varied weather, seasons, and locations. In addition, manual annotation needs expert knowledge of a specific field and a lot of time and labor. Thus, the collection of a large-scale well-annotated dataset costs a lot. In this way, both the source and target datasets are inaccessible; the insufficient data causes poor domain generalization performance. For this reason, a cheap and

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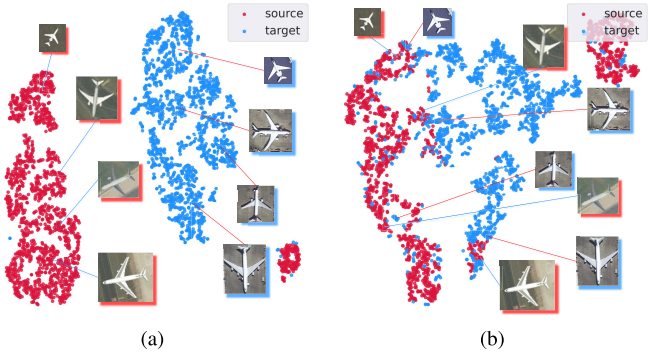

Fig. 1. Diagram of the domain adaptation. We use the tSNE [1] to visualize the instance-level feature vectors for the semisynthetic source (red) and real-world target (blue) domain. (a) Detector trained on the semisynthetic dataset shows an unsatisfactory performance on the real-world dataset; the main reason is that the cross-domain features are not aligned. (b) After the cross-domain feature alignment, the domain gap is reduced, and the accuracy is improved. Better view in color.

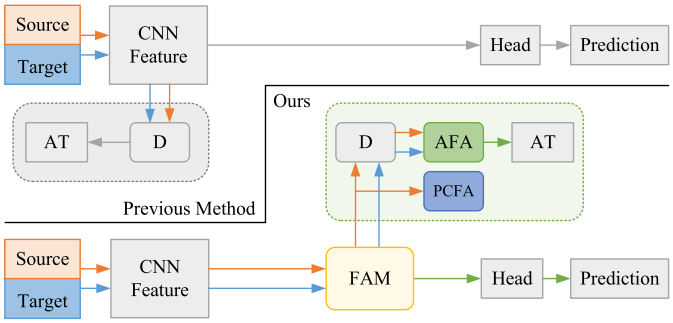

Fig. 2. (Top) Domain adaptation pipeline of the previous detector. (Bottom) Proposed FADA. The previous method aligns the convolutional feature by the adversarial training (AT) on a domain discriminator (D). Our detector leverages the feature alignment module (FAM) to align features in spatial to get the instance-level features and then conducts the domain alignment on the selected background and instance-level foreground features by the proposed AFA and PCFA.

efficient solution with a low cost to automatically acquire samples is desired.

2) Alignment of Confusing Feature: Unlike the nature scene, the remote sensing images have such complicated scenes that detectors are confused to discriminate the targets from the background and cause a low accuracy. The features of the confusing background (bg) and ambiguous foreground (fg) are incorrectly aligned and inseparable in the source domain, such as the airplane in Fig. 1. The present methods align the cross-domain features in an adversarial way; they can hardly deal with the confusing feature within each domain separately. Thus, these source domain features are directly aligned with that of the target domain; this problem is aggravated on the target dataset. Based on this insight, the domain adaptive object detector should focus on aligning the confusing bg/fg features of the cross-domain aerial imagery.

This article tackles these problems from these two aspects as well. For the data acquirement, we propose a semisynthetic solution to rapidly and automatically generate remote sensing datasets. It can relieve the dependence on the expensive real-world source dataset. For the feature alignment, we propose a feature-aligned domain adaptive object detector (FADA). It can promote the unsupervised domain adaptation from the semisynthetic domain to the real-world domain and further alleviate the dependence on the annotations of the target datasets.

To acquire the source dataset at a low cost, we use the computer graphic 3-D engine to build up a semisynthetic data generator (SDG). Specifically, we combine the real-world remote sensing background imagery with the synthetic objects in the 3-D modeling engine [25] and take photos in a top-down orthographic view by a virtual camera. In this way, we build up an automatic SDG. This tool can rapidly generate tens of thousands of samples with automatically well-annotated labels. Meanwhile, the imagery and objects can be replaced according to a specific task, such as airplane detection in the airport, vehicle detection in the car park or on the road, and ship detection in open waters. The generated dataset can be used in the domain adaptive object detection task of many remote sensing scenes.

For the alignment of confusing features, we propose to align foreground features across domains and align the ambiguous background feature within the source domain. Based on our insight, we present the AFA and the PCFA to tackle this problem; the pipeline comparison with the previous method is illustrated in Fig. 2. The AFA chooses the foreground features from the source domain and then aligns them with the target domain by the AT. This module can effectively improve the adaptation accuracy on the target domain by a large margin. The PCFA builds up the prototype for each category and chooses the confusing feature with poor confidence score both for the foreground and background, and then explicitly aligns them with the corresponding prototypes, such as contrastive learning. These two modules compensate for each other to mitigate the domain gap of the foreground features and suppress the incorrect alignment of the confusing features.

The main contributions of this article are given as follows.

1) A semisynthetic solution is proposed to automatically generate remote sensing datasets with annotation; it can replace the expensive real-world source dataset.
2) An FADA is proposed to promote the unsupervised domain adaptation from the semisynthetic to the real-world remote sensing dataset; it can further alleviate the dependence on the annotations of the target datasets.
3) For the feature adaptation, an AFA is proposed. It can bridge the domain gap of foreground features across domains and significantly improve the adaptation performance.
4) For the alignment of confusing features, a PCFA is proposed. It can further align the foreground feature and suppress the incorrect alignment of the confusing features.

To verify the validity of the proposed method, we carry out extensive domain adaptation experiments on the remote sensing object detection across different domains, including our proposed semisynthetic airplane and vehicle detection dataset, the public DOTA [13], xView [26], and the CAR ParKing lot dataset (CARPK) [27] dataset. We implement the domain adaptation on the popular one-stage detector RetinaNet [7] under the proposed FADA framework. The adaptation from the synthetic airplane to DOTA [13] shows that the average

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precision (AP) is significantly improved by 15.7%, and the adaptation from xView [26] to DOTA [13] also shows more than 4.7% mean AP (mAP) improvement. The experiment results indicate that the proposed domain adaptation method can effectively align the cross-domain feature to bridge the domain gap and significantly improve the generalization performance on the target domain in an unsupervised way.

To our best knowledge, this study is the first work to transfer object detection knowledge from the semisynthetic dataset to the real-world remote sensing dataset and build up the first standard benchmark. We hope this work would contribute to the related research in the remote sensing community.

II. Related Works

In the past few years, domain adaptation has been extensively studied in the field of image classification, semantic segmentation, and object detection. However, the published articles mainly focus on the nature scene, while few researchers pay attention to the remote sensing field, especially for the domain adaptive object detection. The main reason lies in two parts: 1) the public dataset and standard benchmark are rare and 2) the domain adaptive object detection has been developed for only a few years; the adaptation performance is challenged by the complicated remote sensing scene.

A. Synthetic Dataset

The synthetic dataset is usually produced by the computer rather than captured by the real sensor in a nature scene. Benefitting from the cheap and convenient acquisition method, synthesis has been adopted into many fields, including semantic segmentation and object detection. The study of GTA5 [28] and SYNTHIA [29] build up a fully synthetic dataset for semantic segmentation; the study of [30] builds up a synthetic crowd counting dataset. These works show that the synthetic dataset has promising generalization and, hence, can free humans from heavy data annotations [30]. The study of virtual Karlsruhe Institute of Technology and Toyota Technological Institute (KITTI) [31] and Sim10K [32] construct the synthetic dataset to imitate the real-word object detection dataset KITTI [33]; the foggy-cityscapes [34] synthesizes the foggy weather by algorithm on the usual images. These datasets are widely studied, and the stand benchmark is continually updated.

Recently, the study of [35] investigates the generalized semantic segmentation performance of the adaptation from synthetic game data to real-world remote sensing images. They leverage the 3-D game engine to collect the synthetic satellite imagery rendered from aerial view. A CycleGAN [36] is implemented to mitigate the domain gap between the "game domain" and the "real-world domain." However, this study is limited by the unique geographical location and the corresponding map of the game engine. Hence, it is unable to be adopted into other scenes or other categories for semantic segmentation tasks.

Different from these articles, our work can construct the semisynthetic object detection dataset in remote sensing scenes. Our solution is to combine the real-world remote sensing background imagery and the 3-D synthetic object. It is flexible to be adopted into more scenes by simply replacing the background imagery and the synthetic objects.

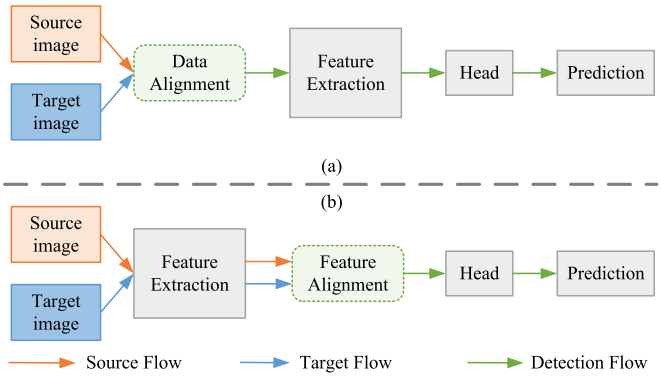

Fig. 3. Different pipelines of the domain alignment methods. (a) Reconstruction-based method directly aligns the cross-domain images to explicitly bridge the domain gap. (b) Adversarial-based methods implicitly align the cross-domain features by AT to narrow the cross-domain semantic gap. Our FADA is based on the second pipeline.

B. Cross-Domain Object Detection

In recent years, more and more domain adaptive object detectors have been developed to address the cross-domain feature misalignment problem. These algorithms can be roughly organized into three types according to the technology roadmap: 1) reconstruction-based methods; 2) adversarial-based methods; and 3) others. The first two methods are commonly used in an unsupervised way, so we focus on these methods. The pipelines of them are compared in Fig. 3.

1) Reconstruction-Based Methods: Reconstruction-based methods bridge the domain gap on images. Some detectors [37]–[39] use the CycleGAN [36] to transfer the source images into the target domain [37] or the middle domain [40]. This method can directly mitigate the domain gap by changing the data distribution. Following this pipeline, some works [37], [38] also contain the feature adaptation in the next step. Hence, they are classified into the "multistep" adaptation methods in some articles [41]. With the reconstruction by CycleGAN, the transferred fake target images have corresponding labels. Thus, the detector can be trained in a supervised way [39] and has a better performance on the target domain. This method can be easily adapted to many tasks, such as the translation from synthetic domain to real-word domain [30], [37], [42], between the day-time domain and the night-time domain [39], [43], and between the thermal domain and the RGB domain [44]–[47].

2) Adversarial-Based Methods: Adversarial-based methods bridge the domain gap on features. Inspired by the methodology of the "domain-adversarial training" [48], the DA-faster [20] implements the AT into the task of the domain adaptive object detection and achieves promising performance. This method connects the deep feature extractor and a domain discriminator by the gradient reversal layer (GRL) [48]. By jointly training the object detector and the domain discriminator, the feature extractor can learn to extract the domain-agnostic feature to confuse the domain discriminator. In this way, the cross-domain features are implicitly aligned. Following this pipeline, many detectors are proposed, such as the strong-weak (SW)-faster [23], multi-adversarial (MA)-faster

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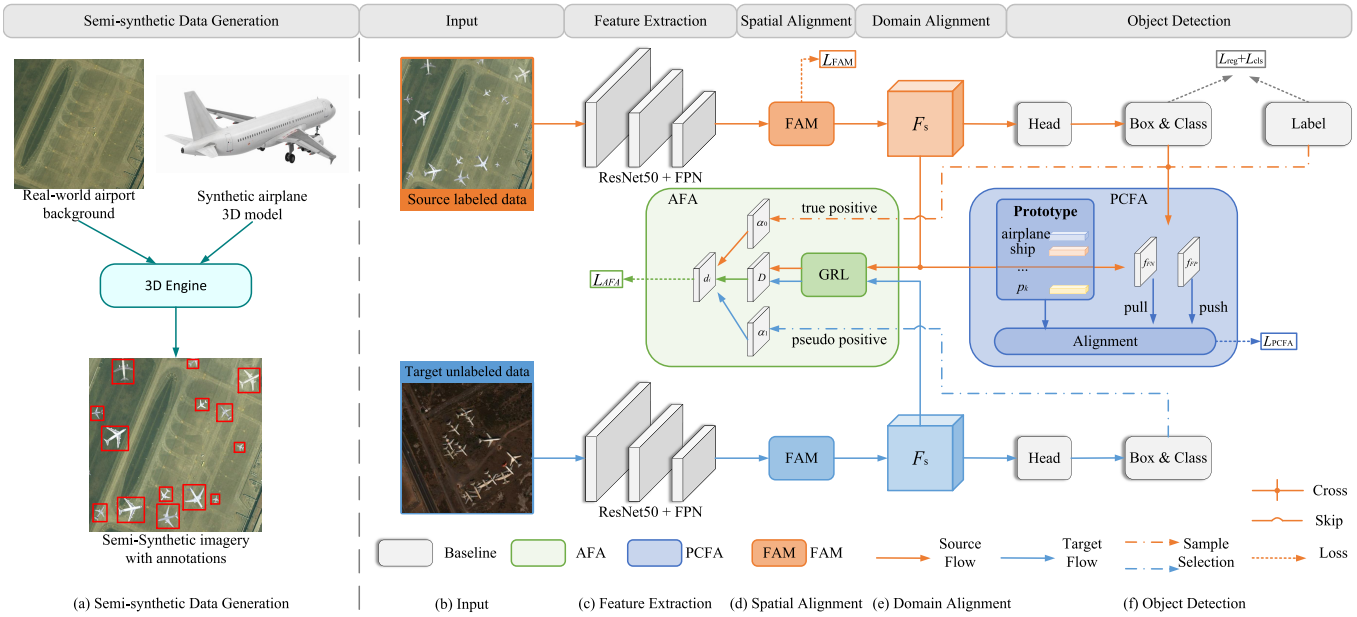

Fig. 4. Overall framework of our proposed method. (a) First, we build up an SDG in the 3-D engine to rapidly produce a large number of semisynthetic images with automatic annotation by computer. (b) Then, these labeled semisynthetic images are taken as the source domain data to transfer to the real-world remote sensing images. (c) Base detector extracts feature from the input. (d) Third, the FAM gets the instance-level features by spatial alignment. (e) AFA adaptively aligns the cross-domain foreground instance feature. The PCFA rectifies the misaligned confusing background feature and foreground instance feature by the prototype alignment. (f) Finally, the aligned features are fed into the detection head to predict the bounding box and category confidence.

[49], adapting object detectors via Selective Cross-Domain Alignment (SCDA) [50], and Hierarchical Transferability Calibration Network (HTCN) [40]. The latest methods also inherit this scheme and achieve impressive performance [51]–[53].

In the remote sensing field, the study of [54] proposes a domain adaptive vehicle detector to promote the generalized performance in the target domain. This work implements the correlation alignment (CORAL) and the adversarial-based domain adaptation to the vehicle detector and achieves promising results. Despite the improvement, the detector is merely trained on the source domain yet has high accuracy on the target domain; hence, the domain gap is not challenging enough. In addition, the source domain also needs a large number of manual annotations.

Following these works, we propose a domain adaptive object detector following the adversarial-based methods. Different from these solutions, it can mitigate the domain gap for the sparse foreground remote sensing object and the complex background, respectively. Besides, the proposed semisynthetic sample generator can replace the real-world source-domain dataset to train this detector and, hence, can significantly reduce the demand for expensive manual annotations. The methodology is detailed in Section III.

III. Methodology

In this work, we propose an SDG to automatically generate the remote sensing dataset and propose an FADA to alleviate the cross-domain feature misalignment in the domain adaptation from the semisynthetic dataset to the real-world dataset. The architecture is illustrated in Fig. 4. The proposed components can be elaborated on as follows.

1) *SDG:* Generating annotated semisynthetic samples for the remote sensing object detection automatically.
2) *AFA:* Aligning the cross-domain foreground feature implicitly.
3) *PCFA:* Suppressing the incorrect alignment of the confusing fg/bg features explicitly.

A. Preliminary Definition

The goal of domain adaptation is to transfer the extracted knowledge from the source dataset to the target dataset. Generally, the distribution of these two datasets is similar but not exactly aligned; hence, the direct transfer without cross-domain alignment will lead to low generalization performance. The domain gap can be caused by the different image sensors, capture time, weather, data source, platform, illumination, and so on.

Following the previous unsupervised domain adaptation (UDA) works [20], [23], [24], [40], [49], [50], [54], [55], our proposed method generalizes the detector from the annotated source domain to the unlabeled target domain as well. Formally, the dataset from the source domain $\mathcal{D}_s$ has a large number of labeled samples $D_s = \{(x_i{}^s, y_i{}^s)\}$, and the dataset collected from the target domain $\mathcal{D}_t$ only has the unlabeled images $D_t = \{(x_j{}^t)\}$. These samples are drawn from the similar but not aligned distributions $P(x_s, y_s)$ and $Q(x_t, y_t)$. The misalignment between P and Q is generally named the domain gap or domain shift $\mathcal{D}_\Delta$. The domain adaptation aims to mitigate $\mathcal{D}_\Delta$.

B. Framework

Unlike the present popular methods, the FADA is a one-stage detector. It is different from the popular domain adaptive two-stage object detectors [20], [23], [24], which has the explicit instance-level spatial alignment by the ROI-Align [56] and the further cross-domain feature alignment for each

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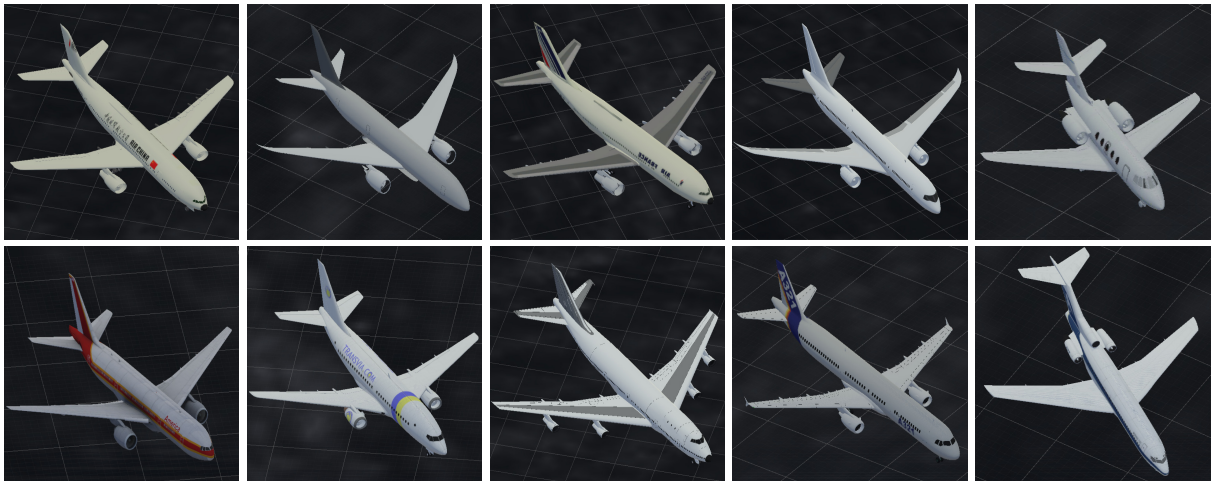

Fig. 5. Illustration of the synthetic 3-D airplane models. The 3-D models are publicly accessible and collected from the internet.

ROI [20], [56]. The fully convolutional one-stage detector is unable to capture the spatially aligned instance-level feature [57], such that the corresponding domain adaptation is not applicable. However, the instance-level feature adaptation for the foreground object is crucial because the complex remote sensing background is not suitable for the direct domain adaptation. Recently, the study of feature Aligned Single-Shot Detection (ASSD) proposes an FAM that can adaptively align the convolutional feature with the foreground objects in spatial [57]; hence, this method enables the one-stage detector to capture the instance-level feature. For that reason, we follow the ASSD [57] to build up our base detector on the popular RetinaNet [7]. Hence, the instance-level domain adaptation method [20], [49], [58] can be implemented in the FADA as well.

As illustrated in Fig. 4, the FADA is built on the RetinaNet [7]. Following the adversarial-based methods [20], [49], the GRL is attached to the feature map after the FPN [59] of RetinaNet. After that, the widely adopted image-level domain discriminator [20] is connected to this layer. Then, the FAM [57] is attached to each FPN layer to collect the spatially aligned features, and the feature on each positive cell is the instance-level feature. After that, the shared detection head predicts the category and bounding box on these features. Meanwhile, the instance-level features are entered into the proposed AFA to adaptively align the foreground feature with the target domain according to the foreground category predictions. Based on the prediction, the proposed PCFA chooses the confusing instance-level feature to conduct the prototype-based feature alignment.

Different from the normal object detector [7], the proposed and implemented modules are removed in the inference stage. Thus, the proposed method improves the generalization performance, while the architecture and efficiency of the base detector are not affected after training. For that reason, the proposed method is plugin-and-play and, hence, has the potential to be applied to the other one-stage RetinaNet-based detectors [6], [60].

C. Semisynthetic Data Generation

The remote sensing imagery is taken in a top-down orthographic view, and the background changes a lot as the geographic location. The previous study generates fully synthetic images in the nature scene by the 3-D game engine [28], [32] or 3-D modeling engine [31] in the front view. Besides, these fully synthetic images have an extremely large domain gap against the real-world remote sensing imagery, such that they are unsuitable for domain adaptation. Hence, these data collection methods and the corresponding datasets are not applicable in the remote sensing scene.

For that reason, we propose a solution to generate the semisynthetic sample for remote sensing object detection. Specifically, we use the Unity3D engine [25] to create a virtual ground plane and a virtual camera in the 3-D environment. Then, we can modify this 3-D scene according to the specific task. Taking airplane detection as an example, we simply place the real-world remote sensing background imagery on the ground to ensure that the "synthetic versus real" domain gap would not be too large. Then, all kinds of publicly accessible synthetic 3-D models (see Fig. 5) of the airplane are placed in the background; meanwhile, the location, orientation, and amount of them are random. We scale these airplane models according to the spatial resolution of the background imagery to imitate the real-world airplane size. After that, we adjust the ground sampling distance (GSD) of the virtual camera according to the background imagery. Then, we manipulate the virtual camera to capture the orthophoto at a great height to imitate the real-world camera of the satellite. The 3-D engine can easily access the location and posture of each 3-D airplane model, so the annotations of them can be automatically produced. Finally, the captured semisynthetic imagery and the corresponding annotations are automatically collected to build up the semisynthetic airplane dataset (SSAD).

In addition to the semisynthetic satellite imagery, our solution can also generate synthetic samples from the unmanned aerial vehicle (UAV) view. We create a virtual parking lot in the 3-D environment and randomly add some synthetic vehicle models into this scene. Then, we adjust the parameters of the virtual camera according to the real-world camera of the UAV. We manipulate the virtual camera to move randomly in the scene such as the real UAV; meanwhile, this camera captures images from the top view. The annotations can also be automatically generated. Based on this work, we build up a

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synthetic vehicle detection (SVD) dataset from the UAV view for the first time.

With our solution, the large-scale remote sensing object detection dataset can be collected with low cost and high efficiency. In general, labeling a 6.7×5.8 km^2 imagery with 200 airplanes costs each expert about 2.5 h. In our synthetic scene, the imagery with the same size and more than 300 synthetic airplanes costs only 1–2 min without any manual labor. To reduce the large-scale real-world dataset demand, we can pretrain a detector on these semisynthetic datasets and fine-tune it on the real-world dataset. In addition, these datasets can also be taken as the source domain in the domain adaptive object detection task; it can avoid the expensive annotation in the source domain.

D. Adversarial-Based Foreground Alignment

As mentioned in Section II-B2, the adversarial-based training is usually implemented to align cross-domain features, such as the image-level alignment in the DA-faster [20]. However, this method directly aligns features without the consideration of the transferability of features. In the remote sensing scene, the foreground objects are sparse, and the amount is small, which means that the background region takes a significant proportion. Hence, this method may cause much useless alignment and even brings negative adaptation. For example, the alignment between the airplane and the vegetation in the forest is not expected; it can dominate the overall adaptation training. To this end, the adaptation of the foreground objects should be emphasized, and the instance-level foreground feature should be explicitly aligned.

Based on this insight, we proposed the AFA for the remote sensing object. As the previous alignment [20], the AFA is an unsupervised training method to mitigate the domain shift. Different from the previous method, the AFA can adaptively choose the foreground samples instead of all the samples.

Specifically, a fully convolutional domain classifier D is connected to the spatially aligned feature maps of FAM by a GRL. The domain classifier D predicts the input domain; the adversarial learning can align cross-domain features by training this classifier. To align the foreground targets from different domains, the loss of the classifier is adaptively weighted by a scale factor α_0 and α_1 in the source and target domains separately. The factor α_0 is the maximal foreground category score $s^{(u,v)}$ of each positive feature cell (u, v), noting that the positive feature cell is decided by the assigned ground-truth label $\mathrm{GT}_{\mathrm{cat}}^{(u,v)}$. In the target domain, the annotation is inaccessible, so the foreground sample is selected by the top foreground score by a threshold T for all the cells. The factor is calculated as follows:

$$\alpha_0^{(u,v)} = \begin{cases} 0, & \mathrm{GT}_{\mathrm{cat}}^{(u,v)} \in \mathrm{neg} \\ s^{(u,v)}, & \mathrm{GT}_{\mathrm{cat}}^{(u,v)} \in \mathrm{pos} \end{cases} \tag{1}$$

$$\alpha_1^{(u,v)} = \begin{cases} 0, & s^{(u,v)} \in [0, T) \\ s^{(u,v)}, & s^{(u,v)} \in [T, 1]. \end{cases} \tag{2}$$

The overall process can be formulated as follows:

$$\mathcal{L}_{\mathrm{AFA}}^{s} = -\frac{1}{\sum \alpha_0} \sum_{i,u,v} \alpha_0^{(i,u,v)} (1 - D_i) \log\left(1 - d_i^{(u,v)}\right) \tag{3}$$

Algorithm 1 AFA: Aligning the Foreground Feature of the Source Domain and the Target Domain

Input: x is the feature map from FPN for D_s and D_t, y^s is the groundtruth label of D_s, D_i is the domain label, σ is the sigmoid activation function, T is a score threshold.

Output: $\mathcal{L}_{AFA}$ is the total loss of AFA.

$x_a = \mathrm{FAM}(x)$
$Det_{box}, Det_{cat} = \mathrm{HEAD}(x_a)$
$d_i = D_1(x_a)$

$GT_{cat}^{s} = \mathrm{LABELASSIGN}(y^s)$
$\alpha_0 = (GT_{cat}^{s} > 0) \cdot \sigma(Det_{cat}^{s})$
$\alpha_1 = (\sigma(Det_{cat}^{t}) \geq T) \cdot \sigma(Det_{cat}^{t})$
$\mathcal{L}_{AFA} = \mathcal{L}_{AFA}^{s}(\alpha_0, D_i^s, d_i^s) + \mathcal{L}_{AFA}^{t}(\alpha_1, D_i^t, d_i^t)$
return $\mathcal{L}_{AFA}$

$$\mathcal{L}_{\mathrm{AFA}}^{t} = -\frac{1}{\sum \alpha_1} \sum_{i,u,v} \alpha_1^{(i,u,v)} D_i \log d_i^{(u,v)} \tag{4}$$

$$\mathcal{L}_{\mathrm{AFA}} = \mathcal{L}_{\mathrm{AFA}}^{s} + \mathcal{L}_{\mathrm{AFA}}^{t} \tag{5}$$

where D_i is the ground-truth domain indicator, the ith image is from the source/target domain when $D_i = 0/1$, d_i is the corresponding domain prediction, and $\mathcal{L}_{\mathrm{AFA}}^{s}$ and $\mathcal{L}_{\mathrm{AFA}}^{t}$ are the foreground alignment losses of the source and target domains separately.

The AFA is optimized in an adversarial way. When $\mathcal{L}_{\mathrm{AFA}}$ is minimized, the domain classifier D learns to discriminate the input domain. At the same time, the GRL reverses the gradient to maximize the loss and updates the feature extractor to confuse this classifier D. When the classifier cannot discriminate the input domain correctly, the cross-domain features are aligned. The detailed steps are described in Algorithm 1.

E. Prototype-Based Confusing Feature Alignment

In remote sensing object detection, the confusing background and foreground are fundamental challenges. In the standard computer vision nature scene, the misclassification of similar categories (cat and dog, bicycle, and motorbike) significantly affects the overall performance due to the small interclass distance. Different from this point, the detector is generally unlikely to misclassify remote sensing objects. However, the confusing background and foreground make it difficult for the detector to correctly discriminate the sparse objects from the extensive background. With the adaptation, this problem also shows in the source and target domains simultaneously. To promote generalized accuracy, the incorrect alignment of the confusing feature, hence, should be suppressed. In the proposed AFA, the domain adaptation of the foreground only focuses on the positive detection, so the false positive detections of the target domain are included, and the false-negative detections of the source domain are filtered out. As a result, the transferred knowledge is also biased for the false detections in the target domain.

Based on this insight, the PCFA is proposed to explicitly suppress the incorrect alignment of the confusing features. As depicted in Fig. 4, the PCFA is connected to the feature

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layer after the FAM. The FAM can capture the spatially aligned feature of the foreground objects [57] so that the PCFA can directly work on the inappropriate instance-level domain alignment. The confusing instance-level foreground feature could cause the false negative detection, while the confusing background feature causes the false positive detection. For that reason, the PCFA contains two parts to suppress them separately.

The proposed PCFA deals with the alignment of the confusing feature by the prototype. The recent work [53] builds up a prototype for the background and foreground feature, and then, the contrastive align loss [53] is implemented to align the region proposal network (RPN) feature with these two prototypes. This method inspires us to propose the PCFA. Different from this study [53], the remote sensing background is complicated such that the prototype of the background is not applicable. To this end, we build up a prototype for each category and conduct the alignment on the confusing foreground and background, respectively.

1) Prototype for Alignment: We build up the prototype of the: true positive detection for each foreground category in the source domain. With the ground-truth labels, each feature cell f is assigned with a positive/negative mask m. Then, the classification confidence s and regression precision intersection-over-union (IoU) is used to choose the true positive samples. Then, we choose the feature cell of the high-quality detection with high overall precision. To prevent the vast inferior detections from overwhelming the superior results, we use the weighted mean value of the selected features to set up the final prototype p^s. It can be defined as follows:

$$m^{(u,v)} = e^{(s^{(u,v)}+\mathrm{IoU}^{(u,v)}-1)*0.5} \tag{6}$$

$$p^s = \frac{1}{\sum_{\mathrm{pos}} m} \sum_{u,v\in \mathrm{pos}} f^{(u,v)} * m^{(u,v)}. \tag{7}$$

Different from the previous work [53], we build a prototype for each category based on all the source training samples; each instance-level feature is extracted by the FAM then heuristically selected and weighted, while the study [53] explicitly constructs two learnable prototypes, and the ground truth of the source domain and the pseudolabel of the target domain are used to extract the RPN feature to align with the prototype. Besides, they [53] update the prototype in training with a single local sample; we update prototypes after each training epoch with all samples for efficiency and stability.

2) Alignment of the Confusing Foreground: The confusing foreground features are expected to be aligned with the foreground prototype. Based on our study, we find that the features of the false-negative detection are far away from all the cluster centers of each category both in the source and target datasets. That is, these features are misaligned with the corresponding prototype. For that reason, we explicitly enforce the confusing foreground features f close to the prototype p^s to minimize the misalignment. Specifically, the feature cells f^i, which have low classification confidence s^i and positive label, are chosen to conduct the alignment. Following the previous work [53], the misalignment between the confusing feature and the prototype is measured by the cosine distance, namely,

$$\cos(x_1, x_2) = \frac{x_1^T \cdot x_2}{\|x_1\| \cdot \|x_2\|}.$$

Formally, the objective is presented as follows:

$$\mathcal{L}_{\mathrm{fg}}^i = 1 - \cos(f^i, p^s) \tag{8}$$

$$\mathcal{L}_{\mathrm{fg}} = \frac{1}{N_{\mathrm{FN}}} \sum_{i\in \mathrm{FN}} (1 - s^i)^\gamma \mathcal{L}_{\mathrm{fg}}^i \tag{9}$$

where $\mathcal{L}_{\mathrm{fg}}^i$ is the loss of the ith chosen feature cell, γ is a modulation factor, such as the FocalLoss [7], we set γ to 2 in experiments, and $\mathcal{L}_{\mathrm{fg}}$ is the overall objective function of the alignment of the confusing foreground.

3) Alignment of the Confusing Background: The confusing foreground features are expected to be far away from the foreground prototype. In the remote sensing scene, the background varies widely so that a single prototype is insufficient to represent the distribution of the background features. Based on this point, the background prototype alignment in the previous study [53] is not applicable anymore. To this end, we attempt to align features based on the foreground prototype in the opposite direction. Specifically, the confusing features f^i are pushed away from all the prototypes p^s to prevent the biased feature mapping. f^i is chosen when the detection of the ith feature cell has high classification confidence s^i and a negative label. The cosine distance is adopted as well; the overall process is formatted as follows:

$$\mathcal{L}_{\mathrm{bg}}^i = \frac{1}{k} \sum_{k=0}^{c} \cos\left(f^i, p_k^s\right) \tag{10}$$

$$\mathcal{L}_{\mathrm{bg}} = \frac{1}{N_{\mathrm{FP}}} \sum_{i\in \mathrm{FP}} (s^i)^\gamma \mathcal{L}_{\mathrm{bg}}^i \tag{11}$$

where p_k^s is the prototype of the kth of the total c categories, $\mathcal{L}_{\mathrm{bg}}^i$ is the objective of the ith chosen sample, the modulation factor γ is set to 2, and $\mathcal{L}_{\mathrm{bg}}$ is the overall objective function of the alignment of the confusing background.

4) Overall Alignment Loss: The overall alignment loss of the confusing feature is the sum of the foreground and background alignment losses. Since the annotation of the target domain is inaccessible, the real prototype p^t is unable to build. In the previous study, the predictions are used as pseudolabels to compute p^t. However, the confusing feature causes false detections so that the pseudoprototype is inappropriate to represent the feature center of the target domain.

In our preliminary experiment on the base detector, we evaluate the confusing feature of the foreground and background; we find that the distribution of the source-domain confusing features is close to that of the target domain. Hence, the prototype alignment in the source domain has the potential to be transferred to the target domain.

As a consequence, the prototype alignment of the confusing foreground and background is conducted in the source domain only. The overall alignment loss is shown as follows:

$$\mathcal{L}_{\mathrm{PCFA}} = \lambda \mathcal{L}_{\mathrm{fg}} + \mu \mathcal{L}_{\mathrm{bg}} \tag{12}$$

where λ and μ are the weight factor; we set them to 1.0 in our experiments.

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TABLE I
DATASET COMPARISON OF THE SOURCE AND TARGET DOMAINS

	dataset	class	data source	annotation	bounding box	GSD	image size	#images	#objects
D_s	SSAD	airplane	3D synthetic	auto	obb+hbb	0.81	$3k \times 3k \sim 7k \times 7k$	300	18891
D_t	DOTA	airplane	GoogleEarth+	manual	obb+hbb	$0.1 \sim 4.5$	$800 \times 800 \sim 4k \times 4k$	267	21767
D_s	SVD	vehicle	3D synthetic	auto	obb+hbb	$0.01 \sim 0.1$	1280×720	3000	213392
D_t	CARPK	vehicle	UAV	manual	hbb	$0.01 \sim 0.04$	1280×720	1448	103632
D_s	xView	Pla+Ship+ST	WorldView-3	manual	hbb	0.3	$2.5k \times 2.5k \sim 5k \times 3k$	216	12398
D_t	DOTA	Pla+Ship+ST	GoogleEarth+	manual	obb+hbb	$0.1 \sim 4.5$	$800 \times 800 \sim 4k \times 4k$	792	119127

- 'Pla+Ship+ST': airplane, ship, storage-tank
- 'GoogleEarth+': Google Earth, Satellite JL-1, GF-2
- 'auto' means automatic annotation, 'obb' means orientated bounding box, 'hbb' means horizontal bounding box.

F. Detection Head

The head of FADA is based on the RetinaNet [7] and ASSD [57] detectors. Following the setting of RetinaNet, two groups of stacked convolutional layers are used to extract the decoupled feature for classification and regression. Differently, the convolutional kernel size is set to 1×1 since the proposed FADA introduces the spatial alignment by FAM for each feature cell [57] to capture the instance-level feature for the foreground.

In addition, the core anchor of ASSD [57] is also implemented, such that each feature cell contains only one core anchor. In this way, the foreground and background feature cells correspond to the positive and negative anchors separately. It can be used to compute (1).

G. Objective Function

The overall training loss of the FADA contains four parts: 1) the detection loss of the base detector; 2) the loss of the spatial alignment of FAM; 3) the loss of the AFA; and 4) the loss of the PCFA.

1) Detection Loss: The detection loss is the same as the RetinaNet [7]. The regression b of the ith positive anchor is used to compute the objective, and the L1 loss is implemented as the criterion. Note that the detection loss is only conducted on the source domain because the annotation of the target domain is inaccessible. The overall formulation is defined as follows:

$$L_{\text{reg}} = \frac{1}{N_{\text{pos}}} \sum_{i \in \text{pos}} \sum_{\theta \in \{x,y,w,h\}} \left| {b_i}^{\theta} - {t_i}^{\theta} \right| \tag{13}$$

$$L_{\text{cls}} = \begin{cases} -\alpha_t (1-p)^{\gamma} \log(p), & \text{if } y = 1 \\ -\alpha_t p^{\gamma} \log(1-p), & \text{otherwise} \end{cases} \tag{14}$$

where b^{θ} and t^{θ} stand for the regression and target of bounding box $\{x, y, w, h\}$, p and y stands for the classification and ground-truth label, and the factors α_t and γ are set to 0.25 and 2.0 following the focal loss of RetinaNet [7].

2) Spatial Alignment Loss: The loss of the spatial alignment of FAM denoted as L_{FAM} has the same formula as (13), but the target t_i^{θ} and prediction b_i^{θ} are encoded and decoded by the core anchor, respectively, rather than the normal anchor group [57].

Totally, the overall training loss of our proposed FADA can be formulated as follows:

$$L = (L_{\text{reg}} + L_{\text{cls}}) + L_{\text{FAM}} + L_{\text{AFA}} + L_{\text{PCFA}}. \tag{15}$$

IV. EXPERIMENTS

In this section, the cross-domain remote sensing datasets are introduced in the first part; the second part details the evaluation metrics and parameter settings of our experiment; the third part presents our experimental results and the corresponding analysis; and the discussion is elaborated on in the last part.

A. Datasets

In this study, we propose a semisynthetic sample generator for the remote sensing object detection; then, we present the FADA detector to transfer the cross-domain knowledge between the real-world and semisynthetic datasets, as well as between different real-world remote sensing datasets. These datasets are compared in Table I and introduced in the following paragraphs.

1) Semisynthetic Airplane Detection (SSAD): In this study, we build up a semisynthetic airplane detection (SSAD) dataset. It can be used to take the place of the source dataset to reduce the demand for the large-scale real-world dataset and expensive annotation. It is noticeable that the background is the real remote sensing imagery from the satellite, and the foreground is the image of the synthetic 3-D model. In the SSAD, more than ten types of airplane models (seen in Fig. 5) are included following the study of A Benchmark Dataset for Fine-grained Object Recognition in High-Resolution Remote Sensing Imagery (FAIR1M) [61], such as the Boeing 737 and Airbus A320. The bounding box annotation of the airplane is the compact oriented rectangle; we convert it into the horizontal box to fit the other dataset (seen in Fig. 6). In principle, the total number of this dataset could be infinite, but we only generate 300 large-scale aerial imageries in our experiments for efficiency. In this dataset, the image size varies from 3k × 3k to 7k × 7k, the GSD is 0.81 m per pixel, and the amount of airplane instances in each imagery randomly varies from 15 to 150. Each image is cropped into the chips with a size of 512 × 512 pixels. All of these images are randomly divided into train and val sets at a ratio of 9:1, and the synthetic train set is taken as the source domain in all experiments.

2) Synthetic Vehicle Detection: As described in Section III-C, we also build up an SVD dataset. Different from the satellite remote sensing dataset, the synthetic images are captured from the UAV view (seen in Fig. 7); we hope that this dataset could contribute to the community. The annotations are the same as the SSAD; the category label is the "vehicle." We only generate 4000 images for the experiment, despite the ability to capture more images.

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(a) Our SSAD (b) DOTA

Fig. 6. (Left) Detection results of our semisynthetic SSAD dataset. (Right) Real-world DOTA dataset. Our solution can automatically produce large-scale imagery with dense objects and arbitrary poses, while the real-world airplane objects in the DOTA dataset are sparse, and the orientation is biased. (a) Our SSAD. (b) DOTA.

Fig. 7. (Top) Detection results of our synthetic SVD dataset. (Bottom) Real-world CARPK dataset. We remove the detection confidence for a clear view.

Each synthetic image has the same size as the image of the real-world camera on the UAV. We crop these images into 720×720 pixels for efficiency. The ratio of the train and val sets is 9:1, and the synthetic train set is used as the source domain in our experiment.

3) CARPK: As discussed above, our SDG can rapidly produce the synthetic dataset and reduce the demand for the real-world dataset. To verify this point, we directly transfer the detector from the synthetic SVD dataset to the real-world CARPK [27] dataset. As illustrated in Fig. 7, the public CARPK dataset [27] is captured by a UAV in the parking lot, and each image is resized into 1280×720 pixels. To inspect the generalization of our SVD, we train the base detector on the SVD and test on the real-world CARPK [27].

4) DOTA: To verify the generalization of the FADA, we also carry out a group of experiments on the DOTA [13] dataset. As shown in Fig. 8, this dataset contains more scenes with large-size and various foreground objects and complex backgrounds. It, hence, can present the data distribution of the real-world remote sensing image. Thus, the DOTA dataset is taken as the target domain in our experiment to judge the generalization performance of our proposed FADA. Limited by the category setting in the source domain, only a part of the common categories of this dataset are used in the task of domain adaptation, and only the horizontal bounding box annotations are used in detection. The FADA is trained and tested on the train set and the publicly accessible val set, respectively, because the involved category subset is not matched with the standard test set on the official evaluation website [13].

5) xView: A group of experiments is conducted on the domain adaptation across different satellites to inspect the effect of our method. Different from the DOTA [13] datasets, which are captured by the Google Earth satellite, the xView [26] dataset is captured by the WorldView-3 satellite. The domain shift between them can be seen in Fig. 8. We also choose the common category with DOTA [13] to conduct the domain adaptation.

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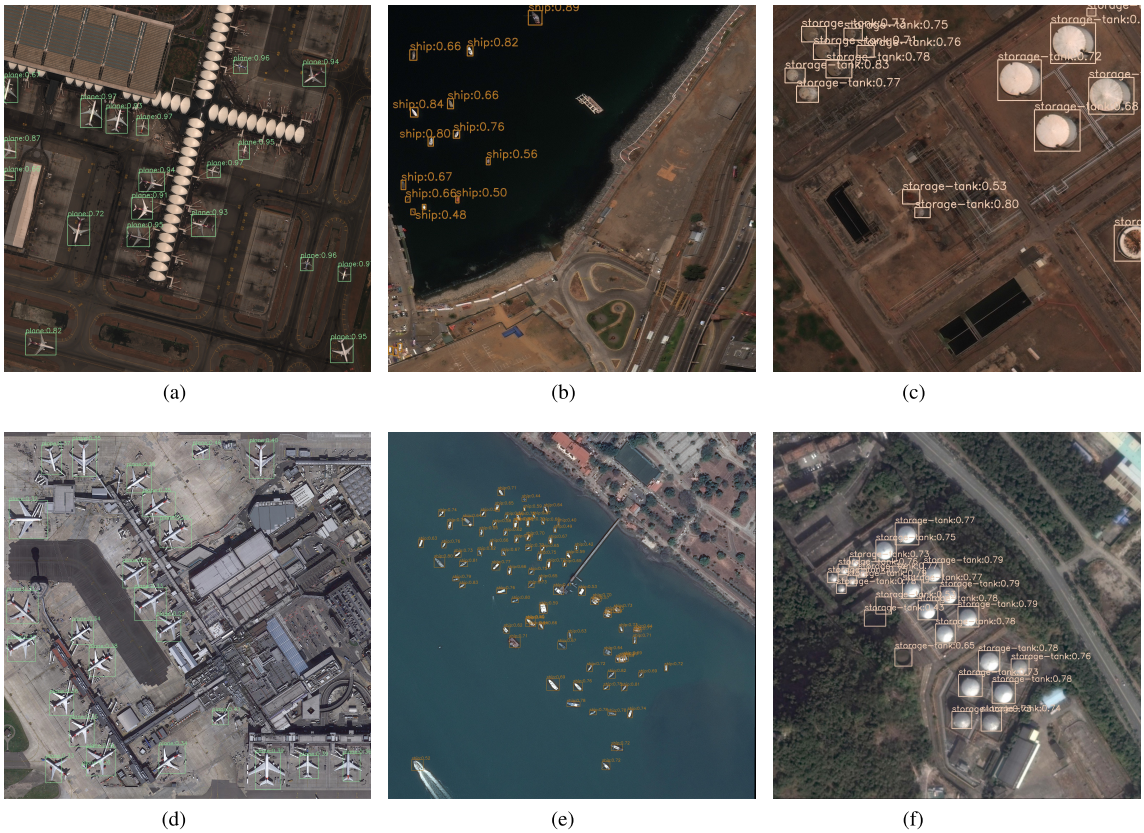

Fig. 8. Detection results and the image comparison between (top) xView and (bottom) DOTA dataset. The images of the DOTA dataset are collected from the Google Earth satellite; the xView is captured by the WorldView-3 satellite. The domain gap mainly comes from the difference in sensors. Better view in color and zoomed-in view. (a) Plane in xView. (b) Ship in xView. (c) Storage-tank in xView. (d) Plane in DOTA. (e) Ship in DOTA. (f) Storage-tank in DOTA.

B. Evaluation Metric

Following the standard object detection task, the normal criteria are also implemented in our experiment to judge the performance of our method. Specifically, the IoU is defined as follows:

$$\text{IoU} = \frac{B \cap B^*}{B \cup B^*} \tag{16}$$

where B is the regressed bounding box and B^* is the corresponding ground-truth box. Generally, the detection box B is taken as a true positive (TP) when IoU ≥ 0.5, false positive (FP) otherwise. B^* is viewed as the false negative (FN) when B^* has no corresponding TP detection.

The accuracy of detection is computed by the classification confidence s_i and the regression precision IoU_i. Each detection is labeled as TP, FP, and FN by a series of confidence and regression threshold values. The regular precision and recall of each category can be calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{17}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{18}$$

The AP of each category is the integral of precision under a series of confidence thresholds. The overall accuracy mAP is formulated as follows:

$$\text{mAP} = \frac{1}{c} \sum_{i=0}^{c} \text{AP}_i. \tag{19}$$

In our experimental results, the mAP_{50} is computed to judge the overall accuracy of the proposed detector.

C. Parameter Settings and Implementation Details

The RetinaNet [7] is the base framework of the FADA, and the FAM of ASSD [57] is inserted after the FPN [59] layer. The backbone is the ResNet-50; we initialize it by the ImageNet [62] pretrained weights. With the core anchor from the ASSD, the anchor setting is slightly different from the RetinaNet. Our preliminary experiment shows that the detector can get higher accuracy when the base scale $S_0 = 2$ due to the small remote sensing objects. The rest of the anchor setting is kept unchanged as the default setting.

Different from the normal detector, the training procedure is divided into two steps. First, we train the base detector on the source dataset to get the source model, denoted as "source" in the following tables. Following the default setting of MMDetection[1] [63], the SGD optimizer is adopted with a momentum of 0.9, and the learning rate is set to 0.02 and is decreased to $2e-3$ and $2e-4$ after the 16th and 22nd epochs, respectively. First, we train the base model for 24 epochs following the default setting of RetinaNet [7]. Second, the trained source model is used to initialize the detector equipped with the proposed adaptation method. Then, we train the FADA for 12 epochs and get the testing accuracy on the target

[1]https://github.com/open-mmlab/mmdetection

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TABLE II
RESULT OF THE ABLATION EXPERIMENTS. THE DETECTOR IS TRANSFERRED FROM THE XVIEW TO THE DOTA DATASET

#	source	RetinaNet	FAM	AT	AFA	PCFA	target	Plane	Ship	Storage-tank	mAP_{50}
1	√	√						53.4	27.3	30.6	37.1
2	√	√	√					62.2	37.5	43.6	47.8 (↑ 10.7)
3	√	√	√	√				62.3	43.4	41.6	49.1 (↑ 10.7 + 1.3)
4	√	√	√		√			**68.8**	42.2	39.6	50.3 (↑ 10.7 + 2.5)
5	√	√	√			√		64.0	42.6	43.9	50.2 (↑ 10.7 + 2.4)
6	√	√	√	√	√	√		66.6	**43.7**	42.4	50.9 (↑ 10.7 + 3.1)
7	√	√	√		√	√		68.0	43.4	**46.2**	**52.5** (↑ 10.7 + 4.7)
8		√					√	89.0	51.2	46.3	62.6

domain. We conduct each experiment on an NVIDIA Tesla V100 GPU.

D. Quantitative Analysis

This study proposes a solution to generate the synthetic dataset in order to relieve the demand for the annotated real-world samples and presents an FADA detector to deal with the cross-domain confusing feature to promote the domain adaptation performance. To inspect the effect of this methodology, we carry out comprehensive experiments on different domains and analyze the results along with the published state-of-the-art (SOTA) methods.

1) Ablation Experiments: The ablation experiments are conducted on the domain adaptation from the xView [26] to the DOTA [13] dataset; we train detectors on the labeled xView train set and the unlabeled train set of DOTA and then test on the DOTA val set. As sketched in Fig. 8, the source and target images are significantly different due to the different sensors of satellites. The objects of the xView are labeled by many fine-grained categories, but the DOTA only has a few superior categories. Thus, we merge the subcategories into the superior category of xView to match the category setting of the target dataset DOTA [13]. For example, the motorboat, sailboat, and fishing vessel are merged into the ship category. Finally, the plane, ship, and storage tank categories are engaged in our experiments.

a) Baseline: First, we train the RetinaNet [7] on the training set of the xView dataset (denoted as "source" in Table II) and directly test on the DOTA testing set. Then, we train the base detector on the DOTA training set (denoted as "target" in Table II) and test on the DOTA dataset. The results are detailed in the first and eighth rows of Table II separately. Compared with the "target" model, the performance of the "source" base detector is unsatisfactory. It is a large performance gap that needs to bridge.

b) FAM: Then, we implement the FAM [57] to improve the base detector. As discussed in the study of ASSD [57], the FAM can adaptively adjust the feature sampling location and the receptive field; thus, the detector has better robustness and generalization for the complex remote sensing scene. Our experimental result also shows a significant generalization accuracy improvement by more than 10% mAP. In addition to the robustness, the overall performance improvement plays an important role. As discussed in a previous study [55], the performance of the source dataset is an essential factor to affect the domain generalization accuracy. The related theory is discussed in Section V-A.

c) AT: For the domain adaptation, we implement the widely adopted AT with the GRL and domain discriminator, such as the DA-faster [20]. The result shows that the overall accuracy is gained by 1.3% mAP. We believe that AT has the potential to bridge the domain gap.

d) AFA and PCFA: Following the AT, we focus the domain adaptation on the foreground features by AFA; it can prevent the vast background from overwhelming the foreground. As a result, the overall performance is improved by more than 2.5% mAP. It can be proof of the effectiveness of the AFA. We argue that the proposed AFA filters out most of the noisy background and makes the feature alignment focus on the transferable features both in the source and target domains. Such that the cross-domain foreground features are aligned better. The following experiment with the proposed PCFA also verifies its effectiveness by 2.4% mAP overall accuracy improvement.

e) Negative effect of AT: Then, we equip the base detector with all the modules to inspect the domain generalization performance. The result shows that the overall accuracy is merely gained by 3.1% mAP. As introduced in the previous sections, the AFA and PCFA are proposed to align the cross-domain confusing foreground and background features. However, the "AT" is conducted on every feature cell of the convolutional feature map. Hence, the "AT" may introduce the negative adaption to damage the overall performance. To this end, we attempt to remove the feature cell-wise "AT" in the last ablation experiment, and then, the detector becomes our proposed FADA. In Table II, the result shows a better improvement by 4.7% mAP. Compared to the base detector, our FADA gets an impressive improvement by 15.4% mAP. It validates our analysis and demonstrates the effectiveness of our method.

2) Generalization of the Synthetic Datasets: In this article, we proposed a semisynthetic solution to tackle the problem of the expensive annotated remote sensing dataset. To inspect the feasibility of this method, a group of experiments is carried out on our synthetic dataset. We merely train the base detector RetinaNet [7] on the SVD train set and then directly test on the real-world CARPK test set. The images of these datasets are collected from the UAV view, while the SVD is thoroughly generated by a computer without any manual labor, and the imagery comparison is shown in Fig. 7. To exploit the generation of the synthetic dataset, we gradually increase the total number of the synthetic samples until the accuracy become stable. The evaluation results are reported in Fig. 9.

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TABLE III

RESULT OF THE DOMAIN ADAPTATION EXPERIMENTS FROM THE SYNTHETIC TO THE DOTA DATASET. THE RESULTS OF SOTA METHODS ARE COMPARED AS WELL

method	source	RetinaNet	FAM	AT	AFA	PCFA	MS-test	target	Recall	AP
other methods										
DAFaster(CVPR'18) [20]	√			√					55.6	53.0
SWFaster(CVPR'19) [23]	√			√					56.1	47.6
HTCN(CVPR'20) [40]	√			√					46.3	40.8
EveryPixelMatters(ECCV'20) [51]	√			√					55.5	46.5
our method										
1	√	√							39.0	11.2
2	√	√	√						71.7	42.8
3	√	√	√	√					68.0	53.7
4	√	√	√		√	√			73.9	60.1
5	√	√	√	√	√	√			74.5	58.5
6	√	√	√		√	√	√		**82.9**	**66.1**
7		√						√	88.0	80.1

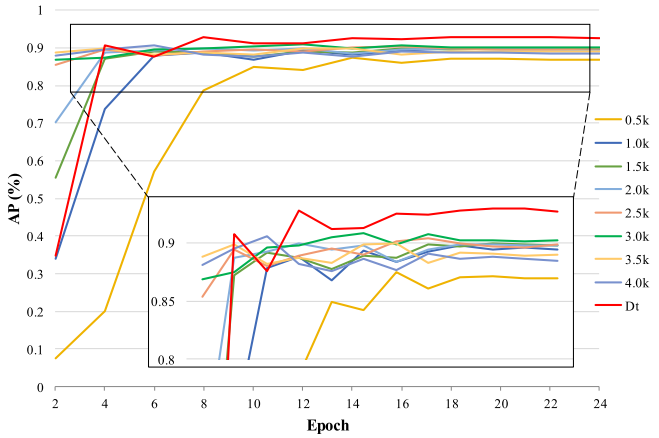

Fig. 9. Generalization performance of the synthetic datasets. We train the base detector on the synthetic SVD with 0.5k ∼ 4k samples and directly test on the CARPK dataset. "D_t" means the reference experiment to train and test on CARPK dataset. Best view in color.

The quantitative results demonstrate that, as the amount of our synthetic samples increases, the generalization performance gradually increases as well. It is because the diversity of samples increases with the scale of the dataset. After the point of 3000 samples, the accuracy increment becomes negligible. It means that the generalization performance of the synthetic datasets reaches the upper boundary. The best accuracy is quite near the "target" model; this result tells the powerful generalization of the synthetic datasets. That is to say, the synthetic dataset has the potential to take the place of the real-world remote sensing dataset in some detection tasks.

However, there is still a performance gap between the synthetic (green curve) and real datasets (red curve). We argue that the domain gap causes this, and it is hard to minimize merely by the data generation and augmentation. For that reason, we conduct a group of domain adaptation experiments to mitigate the domain gap across the synthetic and the real-world remote sensing datasets in the following section.

3) Adaptation From Synthetic to Real-World: In this experiment, we conduct the domain adaptation from the semisynthetic dataset to the real-world remote sensing dataset for the airplane category. Specifically, the detector is trained with the labeled SSAD train set and the unlabeled DOTA train set and tested on the real-world DOTA val set. Unlike the previous vehicle detection in a fixed parking-lot scene, the SSAD dataset has much more varied background scenes and much sparser foreground in addition to the significant domain shift. Thus, compared with the "target" model, the base "source" detector has an inferior generalization performance of about 11.2% mAP.

In Fig. 10(a), we draw the feature distribution of the source and target domains by the t-SNE [1] for clear comprehension. In this figure, the source and target features are misaligned, which indicates that the domain gap needs to bridge. It is the domain gap that causes the cross-domain generalization performance gap. To this end, we add the proposed method to the base detector to enhance the performance on the target domain. Note that no one annotation of the DOTA dataset is used in this experiment except for evaluation.

The FAM could help the detector filter the background and extract each airplane's foreground feature and enhance the robustness. Thus, the generalization accuracy on the target domain is increased from 11.2% mAP to 42.8% mAP. Then, regular AT is implemented to bridge the domain gap directly. The result shows a significant improvement of 10.9% mAP without any cost. Then, the proposed AFA and PCFA are implemented to deal with the confusing feature misalignment. Then, we get an accuracy improvement of 17.3% mAP to achieve the best adaption performance of 60.1% mAP. These quantitative results provide powerful proof of the effectiveness of the proposed method.

At the fifth line of Table III, we add the widely adopted AT to the FADA, and the overall precision is dropped by 1.6% mAP. In our view, this is because the overall domain alignment may include the diverse backgrounds unfit to transfer. Finally, the multiscale (MS) testing strategy is implemented to suit the diverse spatial resolution of the real-world testing images, and we totally get 66.1% mAP near the target performance. Besides, we also visualize the feature of the FADA in Fig. 10. It shows that the domain gap is effectively mitigated, and cross-domain features are aligned better than before.

Compared to the SOTA methods [20], [23], [40], [51], our FADA achieves a competitive performance in Table III. The representative framework DA-faster [20] achieves 53.0% AP, and its successor SW-faster [23] achieves 47.6% mAP.

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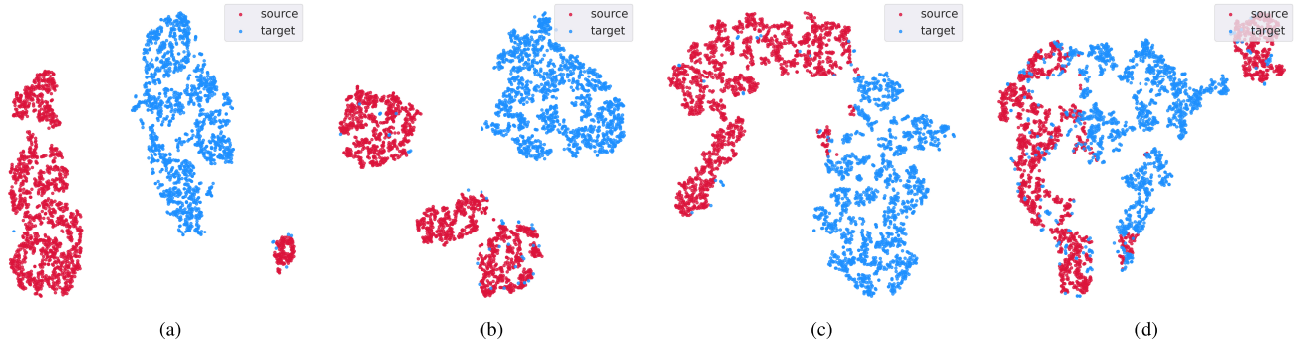

(a) (b) (c) (d)

Fig. 10. Visualization of the cross-domain (semisynthetic SSAD→ real-world DOTA) feature alignment process. The instance-level features are visualized by t-SNE [1]. (a) Misalignment between the source (red points) and target (blue points) domains. (b)–(d) Our aligned results after different training epochs. The adaptation accuracy gradually increases as the cross-domain features are aligned step by step. Better view in color. (a) Source, AP=11.2%. (b) Epoch = 1, AP = 36.9%. (c) Epoch = 4, AP = 52.5%. (d) Epoch = 13, AP = 58.8%.

TABLE IV

COMPARISON OF SOTA METHODS. THE DETECTOR IS TRANSFERRED FROM THE XVIEW TO THE DOTA DATASET

method	source	target	Plane	Ship	Storage-tank	mAP_{50}
other methods						
DAFaster(CVPR'18) [20]	√		47.5	24.9	22.0	31.5
SWFaster(CVPR'19) [23]	√		57.3	30.0	34.6	40.6
HTCN(CVPR'20) [40]	√		54.7	30.9	32.2	39.3
EveryPixelMatters(ECCV'20) [51]	√		48.8	38.3	42.0	43.0
our method						
source	√		53.4	27.3	30.6	37.1
FADA	√		**68.0**	**43.4**	**46.2**	**52.5**
target		√	89.0	51.2	46.3	62.6

The recent method HTCN [40] achieves 40.8% mAP. For the one-stage detector, the SOTA method EveryPixelMatters [51] is based on the anchor-free fully convolutional one-stage object detection (FCOS) [6]; it gets 46.5% mAP. Our method achieves higher accuracy than the SOTA methods in the domain adaptation of airplane detection from the semisynthetic SSAD to the real-world DOTA.

4) Adaptation Between Real-World Satellites: In Table IV, we investigate the FADA along with the SOTA methods on the domain adaptation between the real-world remote sensing datasets from different satellites. Specifically, the source domain is the xView [26] collected by the WorldView-3 satellite; the target domain is the DOTA [13] collected from the Google Earth and GaoFen satellite; and the images are illustrated in Fig. 8. The detector is trained with the labeled xView train set and the unlabeled DOTA train set, evaluated on the DOTA val set. Unlike the adaptation from the semisynthetic to the real-world dataset, the domain gap is much smaller in this experiment. However, the task is much more challenging as well because the diversity of these two real-world datasets is more significant.

The results exhibit that our FADA realizes the best performance. Since the related works are published in standard computer vision, there is no public benchmark to analyze. Thus, we reimplement some of the open-source methods on the domain adaptation from xView to DOTA with their official default parameter setting. Specifically, the current SOTA method HTCN[2] [40] gets an adaptation performance of about 39.3% mAP. The latest method EveryPixelMatters[3] [51] achieves the top accuracy of 43.0% mAP. The early methods, such as DA-Faster[4] [20] and SW-Faster[5] [23], show lower performance. Finally, our proposed FADA achieves the SOTA overall accuracy in this experiment.

To the best of our knowledge, we report the first benchmark of the domain adaptive object detection in the remote sensing field; we hope that it contributes to the research of this community. Besides, the related source and target datasets are publicly accessible; researchers can directly compare their study with these results.

E. Qualitative Analysis

Fig. 10 compares the feature distribution of the semisynthetic source and the real-world target dataset by the t-SNE [1]. The misaligned feature centers indicate the domain gap between these two datasets. The domain adaptation aims to mitigate the domain gap. Then, Fig. 10 shows our aligned results. Compared with the base result, the feature centers of these two datasets are aligned better. Meanwhile, the overall accuracy is improved by a large margin. It further confirms that the domain gap is one of the reasons for the generalization performance gap, and the alignment of the cross-domain feature is able to alleviate this problem.

V. DISCUSSION

A. Explanation From the Theoretical Insight

In the study of the theory of domain adaptation [55], the expected error on the target domain is bounded as follows:

$$\forall h \in \mathcal{H}, \quad R_{\mathcal{T}} \leq R_{\mathcal{S}}(h) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{S}, \mathcal{T}) + C \qquad (20)$$

[2]https://github.com/chaoqichen/HTCN

[3]https://github.com/chengchunhsu/EveryPixelMatters

[4]https://github.com/krumo/Domain-Adaptive-Faster-RCNN-PyTorch

[5]https://github.com/VisionLearningGroup/DA_Detection

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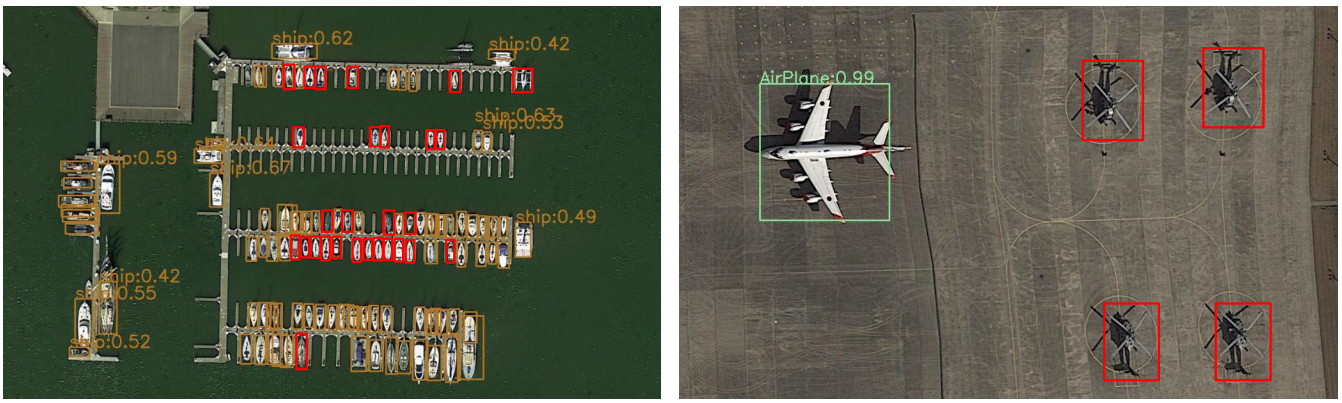

(a) (b)

Fig. 11. False detection results on the DOTA dataset. (a) Distribution of the densely packed ship is different from the sparse arrangement in the source domain xView. (b) "Helicopter" is an unknown aircraft for the source domain. Only a part of the labels is presented for a clear view.

where h denotes the hypothesis for the domain adaptation, $R_{\mathcal{S}}$ and $R_{\mathcal{T}}$ stand for the expected error on the source and target domains, $d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{S},\mathcal{T})$ means the domain divergence, and C is the error of the ideal hypothesis [55]. Domain adaptation needs to find an optimal hypothesis with minimal $R_{\mathcal{T}}$. The right-hand side of this equation decides the upper error bound of the model.

According to the study of FAM [57], this module could promote the performance of the base detector, and our source model is also improved, namely, the $R_{\mathcal{S}}(h)$ is reduced. With the AFA, the domain diversity of the foreground is mitigated by the AT, which means that $d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{S},\mathcal{T})$ gets smaller. Finally, the PCFA aligns the confusing feature and enhances the discriminative feature to reduce C. In total, the proposed method promotes cross-domain generalization performance by lowering the upper error bound in theory.

B. Error Analysis

The results in Tables II and III show a promising performance improvement, but the transferred model still has a negligible performance gap. Based on our analysis, we think that it comes from two parts.

The first part is the domain divergence. Fig. 11(a) shows the false detection results of the ship in the target domain. In the source domain (xView), the ships are sparse and distributed evenly, as shown in Fig. 8(b). However, in the target domain (DOTA), many small ships are densely packed in the harbor. The proposed method can only deal with the semantic-level cross-domain feature misalignment, but it cannot deal with the spatial distribution divergence. We will work on this problem in the future.

The second part is the lack of supervised learning. Fig. 11(b) shows the false detection results of the airplane in the target domain. In the source domain (SSAD), the included 3-D airplane models can not cover all the real-world aircraft. Hence the "helicopter" is unknown for the detector, and the detection confidence is relatively low. A simple solution is to find the corresponding 3-D model, but it is still limited in the specific task and scene. A potential method is to introduce the corresponding label under the supervised learning or the few-shot learning to improve detection performance.

VI. Conclusion

In this article, we study the severe annotation dependence and the cross-domain feature alignment in remote sensing object detection. We use the 3-D engine to combine the real-world aerial imagery and the synthetic 3-D model to automatically generate the well-labeled semisynthetic datasets at a low cost. Experiments show that the generated dataset has a promising generalization performance and has the potential to replace the real-world dataset to relieve the dependence of the expensive manually labeled remote sensing dataset. Then, we present a one-stage detector FADA to align the cross-domain features, in which the AFA is proposed to align the cross-domain foreground feature, and the PCFA is proposed to adaptively align the confusing background and foreground features. The experiments of domain adaptation from the synthetic to the real-world dataset illustrate that our method can effectively align the cross-domain features to promote the performance on the target domain. The domain adaptation from xView to the DOTA dataset shows that our method achieves state-of-the-art results. Besides, we build up the first standard benchmark for the domain adaptive object detection in the remote sensing field. The SDG and FADA can be jointly used to alleviate the dependence on the rare and expensive real-world datasets in many remote sensing object detection tasks, such as the detection of the airplane above the ocean background and the detection of the rare helicopter. The FADA can align the cross-domain feature at the semantic level, but it cannot tackle the spatial distribution divergence and the unknown category, and we will work on it in future work.

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